



PIRLS Reading Practice — Year 5

Providence English Private School

Set 2: How Bees Make Honey

Informational Text · 14 Questions · 40 Minutes

Name: _____ Class: _____ Date: _____

How Bees Make Honey

Honey has been enjoyed by people for thousands of years, but few of us stop to think about the incredible amount of work that goes into making a single jar. Behind every spoonful of honey lies a remarkable process carried out by thousands of bees working together.

It all begins with the worker bees, who fly out from the hive in search of flowers. Using their long, tube-shaped tongues, they suck up a sweet liquid called nectar from deep inside each flower. A single bee might visit more than a hundred flowers on one trip, storing the nectar in a special pouch inside its body called a "honey stomach." This is separate from the stomach the bee uses for its own food, so the nectar stays clean and ready for the next stage.

Once a bee's honey stomach is full, it flies back to the hive. There, it passes the nectar, mouth to mouth, to another worker bee inside the hive. This process is repeated several times, with each bee adding special enzymes from its body that begin to change the nectar's chemistry. Slowly, the watery nectar starts to thicken and transform.

The bees then spread the partly finished honey into the six-sided wax cells of the honeycomb. At this stage, the mixture is still far too watery to be called honey — it contains far too much water to be stored safely. If left this way, it would spoil quickly.

To solve this problem, the bees fan their wings rapidly over the open cells, creating a steady breeze that speeds up evaporation. Thousands of bees fanning together can raise quite a noticeable hum, which is one reason a healthy hive always seems to be buzzing. Over time, this fanning removes most of the extra water, leaving behind a thick, sticky, golden liquid — the honey we recognise.

Once the honey has reached the right thickness, the bees seal each cell with a thin layer of wax, like a cap on a jar. This sealed honey can now be stored safely for months, providing food for the colony throughout winter, when flowers are scarce and no nectar can be collected.

Beekeepers who wish to harvest honey for people carefully remove some of these wax-capped honeycomb frames, always leaving enough behind so the bees themselves have food to survive. The wax caps are sliced off, and the frames are spun in a machine that uses spinning force to fling the honey out of the cells, without damaging the comb itself. The honey then flows out, is filtered to remove any bits of wax, and is finally ready to be poured into jars.

It takes an extraordinary amount of effort to produce even a small amount of honey. Scientists estimate that to make just one kilogram of honey, bees must visit around four million flowers and fly a combined distance equal to travelling around the world more than twice. The next time you spread honey on your toast, it's worth remembering the thousands of tiny wings that made it possible.

Questions

1. MULTIPLE CHOICE · RETRIEVE

What is nectar?

- ☐ A type of flower
- ☐ A sweet liquid found inside flowers
- ☐ A kind of bee
- ☐ The wax used to build honeycomb

2. MULTIPLE CHOICE · RETRIEVE

What is a bee's 'honey stomach' used for?

- ☐ Digesting the bee's own food
- ☐ Storing nectar separately from the bee's food stomach
- ☐ Producing wax
- ☐ Making the buzzing sound

3. WRITTEN RESPONSE · INFER

Why do you think it is important that the honey stomach is separate from the bee's regular stomach?

4. MULTIPLE CHOICE · RETRIEVE

According to the passage, why do bees fan their wings over the honeycomb cells?

- ☐ To cool down the hive
- ☐ To keep away predators
- ☐ To speed up evaporation and thicken the nectar
- ☐ To communicate with other bees

5. WRITTEN RESPONSE · INTERPRET & INTEGRATE

Explain, in your own words, why the nectar would spoil if the bees didn't remove the extra water from it.

6. MULTIPLE CHOICE · RETRIEVE

Why do bees seal the honeycomb cells with wax?

- ☐ To hide the honey from other bees
- ☐ So the honey can be stored safely for months
- ☐ To make the honeycomb look neater
- ☐ To stop the honey from being golden

7. MULTIPLE CHOICE · INTERPRET & INTEGRATE

What happens to the honey during winter, according to the passage?

- ☐ Bees stop making it completely
- ☐ It provides food for the colony when flowers are scarce
- ☐ It is given to beekeepers only
- ☐ It turns back into nectar

8. WRITTEN RESPONSE · INTERPRET & INTEGRATE

Why do beekeepers make sure to leave some honey behind when they harvest it?

9. MULTIPLE CHOICE · RETRIEVE

What does the spinning machine do to the honeycomb frames?

- (A) Melts the wax completely
- (B) Uses spinning force to fling the honey out of the cells
- (C) Adds more nectar to the frames
- (D) Crushes the frames to extract wax

10. MULTIPLE CHOICE · RETRIEVE

According to the passage, roughly how many flowers must be visited to make one kilogram of honey?

- (A) A hundred
- (B) A thousand
- (C) Four million
- (D) Ten million

11. WRITTEN RESPONSE · EVALUATE

The author says it's 'worth remembering the thousands of tiny wings' that made honey possible. What point do you think the author is trying to make with this ending?

12. MULTIPLE CHOICE · INTERPRET & INTEGRATE

Which of these best describes the overall structure of this passage?

- (A) A story about one bee's adventure
- (B) An argument for why honey is unhealthy
- (C) A step-by-step explanation of a natural process
- (D) A list of different bee species

13. WRITTEN RESPONSE · EVALUATE

Based on the passage, do you think making honey is an easy or difficult process for bees? Use at least one detail from the text to support your answer.

14. MULTIPLE CHOICE · EVALUATE

What is the main purpose of this passage?

- (A) To persuade readers to become beekeepers
- (B) To entertain readers with a fictional bee character
- (C) To inform readers about the process of how honey is made
- (D) To compare different types of honey

1. B — A sweet liquid found inside flowers

The passage describes nectar as 'a sweet liquid' that bees collect from inside flowers.

2. B — Storing nectar separately from the bee's food stomach

The passage explains this is a separate pouch used just for storing nectar, keeping it clean.

3. Written response

Model answer: It's likely important because it keeps the nectar clean and unmixed with the bee's digested food, so the nectar can later be safely turned into honey that other bees and people can eat.

Marking guidance: Full credit: connects separation to cleanliness/safety of the nectar for later use. Partial credit: states it's separate without explaining why that matters.

4. C — To speed up evaporation and thicken the nectar

The text states the fanning speeds up evaporation, removing water so the nectar thickens into honey.

5. Written response

Model answer: Nectar that still contains a lot of water is more likely to spoil or ferment because the extra moisture allows unwanted bacteria or yeast to grow, so removing the water helps preserve the honey safely for a long time.

Marking guidance: Full credit: connects excess water to spoiling/preservation issues, even in simple terms. Partial credit: repeats that it would spoil without explaining the water connection.

6. B — So the honey can be stored safely for months

The passage states the sealed honey can be stored safely for months as food during winter.

7. B — It provides food for the colony when flowers are scarce

The passage explains stored honey feeds the colony through winter when nectar can't be collected.

8. Written response

Model answer: Beekeepers leave honey behind so that the bees still have enough food to survive, especially through winter when they can't collect more nectar.

Marking guidance: Full credit: explains that bees need the remaining honey as food to survive. Partial credit: says 'so bees don't starve' with no further detail, or vague answer.

9. B — Uses spinning force to fling the honey out of the cells

The passage describes the machine using spinning force to remove honey without damaging the comb.

10. C — Four million

The passage states bees must visit around four million flowers for one kilogram of honey.

11. Written response

Model answer: The author likely wants readers to appreciate the huge, often unnoticed effort bees put into making honey, encouraging more respect or gratitude for something people usually take for granted.

Marking guidance: Full credit: identifies the author's purpose as encouraging appreciation/awareness of the bees' effort. Partial credit: only restates the sentence without interpreting its purpose.

12. C — A step-by-step explanation of a natural process

The passage explains, in order, each stage of how honey is made — a clear step-by-step explanatory structure.

13. Written response

Model answer: Making honey is clearly a difficult, effort-intensive process — the passage notes that producing just one kilogram requires visiting about four million flowers and flying a distance equal to circling the world more than twice.

Marking guidance: Full credit: gives a judgement (difficult/effortful) supported by a specific fact from the passage. Partial credit: gives a judgement without textual support.

14. C — To inform readers about the process of how honey is made

The passage is written to inform and explain a real-world process, not to persuade or tell a story.